

SRINIVAS UNIVERSITY

**INSTITUTE OF COMPUTER SCIENCE & INFORMATION
SCIENCES**

**CITY CAMPUS, PANDESHWAR,
MANGALORE - 575 001**

Question Bank Answers with Bloom's Model

CLOUD COMPUTING

B. C. A - V SEMESTER



**Compiled by
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CLOUD COMPUTING QUESTION BANK

Exam	V Semester	Paper code	20BCASD54/20BCANT55
Subject	Cloud Computing	Class	III BCA
Maximum marks	50	Time	2 Hours

Weightage Table

Sl. No	Objectives	Marks	Percentage marks
1.	Knowledge (Remembering)	05	10
2.	Understanding	20	40
3.	Application	15	30
4.	Skill	10	20
Total		50	100

Blueprint with Unit wise Marks

Unit	Remembering			Understand			Application			Skill			Total
	OT	SA	Unit wise Marks	OT	SA	Unit wise Marks	OT	SA	Unit wise Marks	OT	SA	Unit wise Marks	
1	1(1)	1(4)	5	1(1)	1(4)	5	-	-	-	-	-	-	10
2	-	-	-	2(1)	1(4)	6	-	1(4)	4	-	-	-	10
3	-	-	-	-	1(4)	4	2(1)	1(4)	6	-	-	-	10
4	-	-	-	1(1)	1(4)	5	-	1(4)	4	1(1)	-	1	10
5	-	-	-	-	-	-	1(1)	-	1	1(1)	2(4)	9	10
	05			20			15			10			50

Blueprint

Unit	Remembering		Understand		Application		Skill		Total
	OT	SA	OT	SA	OT	SA	OT	SA	
1	1(1)	1(4)	1(1)	1(4)	-	-	-	-	10
2	-	-	2(1)	1(4)	-	1(4)	-	-	10
3	-	-	-	1(4)	2(1)	1(4)	-	-	10
4	-	-	1(1)	1(4)	-	1(4)	1(1)	-	10
5	-	-	-	-	1(1)	-	1(1)	2(4)	10
	05		20		15		10		50

Unit - I
Multiple Choice Questions
(Questions for Remembering)

1. Which of the following is the second generation of WWW?
 - A. Cloud computing
 - B. Grid computing
 - C. Web 2.0**
 - D. AWS

2. What is example for IaaS?
 - A. Email
 - B. Google App Engine
 - C. Salesforce.com
 - D. Amazon Web Services**

3. Which cloud deployment model crosses isolation and overcomes boundaries?
 - A. Private
 - B. Public
 - C. Hybrid**
 - D. Community

4. What is another name for “Internal cloud”?
 - A. Private**
 - B. Public
 - C. Hybrid
 - D. Community

5. Which of the following deployment model is a issues as “Poor scalability“?
 - A. Private**
 - B. Public
 - C. Hybrid
 - D. Community

6. Which of the following cloud has Complex networking issue?
 - A. Private
 - B. Public
 - C. Hybrid**
 - D. Community

7. Which of the following cloud deployment model Organization having similar computing concern opt for?
- A. Private
 - B. Public
 - C. Hybrid
 - D. Community**
8. SOA stands for:
- A. Software Oriented Architecture
 - B. Service Oriented Architecture**
 - C. Software Originated Architecture
 - D. Service Originated Architecture
9. A Computer network where every computer shares resource of every other computer is?
- A. Cloud computing
 - B. Grid computing**
 - C. Utility computing
 - D. Virtualization
10. Which of the following can turn a computer network into powerful supercomputer.
- A. Cloud computing
 - B. Grid computing**
 - C. Utility computing
 - D. Virtualization
11. Which of the following is not a type of virtualization
- A. Hardware Virtualization
 - B. Server Virtualization
 - C. Storage Virtualization
 - D. Data virtualization**
12. Software as a Service (SaaS) based on open source code is?
- A. Web 2.0
 - B. SOA
 - C. Open SaaS**
 - D. AWS
13. Customer Relationship Management (CRM) is an example of
- A. IaaS application
 - B. SaaS Application**
 - C. PaaS Application
 - D. Utility Application

14. Which of the following offered Grid Engine software that allows engineers at companies to pool the computer cycles.
- A. Apple
 - B. Microsoft
 - C. Sales force
 - D. Sun Microsystems**
15. Which layer of SOA is used to build the functional and technical libraries?
- A. Consumer Interface Layer
 - B. Service Layer
 - C. Service Component Layer**
 - D. Operational Systems Layer

(Questions for Understanding)

16. Cloud in cloud computing refers to _____
- A. Service model
 - B. Internet**
 - C. Service Provider
 - D. Software
17. Organizations are implementing cloud computing for _____
- A. reducing capital expenditure**
 - B. increasing business
 - C. reducing data redundancy
 - D. for user interactivity
18. _____ is services on demand software.
- A. SaaS**
 - B. PaaS
 - C. IaaS
 - D. NaaS
19. Which is also known to be hardware as a service.
- A. SaaS
 - B. PaaS
 - C. IaaS**
 - D. NaaS
20. Which application is well suited for scientific and complex mathematical processing.
- A. Cloud computing
 - B. Grid computing**
 - C. Utility computing
 - D. Open SaaS

21. Single instance of software application services multiple customers, this is seen in?
- A. Grid computing
 - B. Multitenancy architecture**
 - C. SOA
 - D. Open SaaS
22. The ability which allows sharing the physical instance of a single application or resource among multiple organizations or users is _____
- A. Cloud computing
 - B. Grid computing
 - C. Utility computing
 - D. Virtualization**
23. Which layer of SOA have GUI based apps for end users accessing the applications.
- A. Consumer Interface Layer**
 - B. Services Layer
 - C. Service Component Layer
 - D. Operational Systems Layer
24. Developers refer to a set of web services as?
- A. Web 2.0
 - B. Application package
 - C. SaaS solution
 - D. Application program interface (API)**
25. Less vendor lock-in and no licensing is a benefit of
- A. Multi tenant architecture
 - B. SOA
 - C. API
 - D. Open SaaS**

Questions carrying 4 marks

(Questions for Remembering)

1. Explain private cloud mention its advantage and disadvantages.

Private Cloud also termed as “Internal Cloud”; which allows the accessibility of systems and services within a specific boundary or organization. The cloud platform is implemented in a cloud-based secure environment that is guarded by advanced firewalls under the surveillance of the IT department that belongs to a particular organization. Private clouds permit only authorized users, providing the organizations greater control over data and its security. Business organizations that have dynamic, critical, secured, management demand-based requirement should adopt Private Cloud.

Advantages of private cloud are:

- **Highly private and secured:** Private cloud resource sharing is highly secured.
- **Control Oriented:** Private clouds provide more control over its resources than public cloud as it can be accessed within the organization's boundary.

Disadvantages of private cloud are:

- **Poor scalability:** Private type of clouds is scaled within internal limited hosted resources.
- **Costly:** As it provides secured and more features, so it's more expensive than a public cloud.
- **Pricing:** is inflexible; i.e., purchasing new hardware for up-gradation is more costly.
- **Restriction:** It can be accessed locally within an organization and is difficult to expose globally.

2. Explain public cloud mention its advantage and disadvantages.

Public Cloud is a type of cloud hosting that allows the accessibility of systems & its services to its clients/users easily. Some of the examples of those companies which provide public cloud facilities are IBM, Google, Amazon, Microsoft, etc. This cloud service is open for use. This type of cloud computing is a true specimen of cloud hosting where the service providers render services to various clients. From the technical point of view, there is the least difference between private clouds and public clouds along with the structural design. Only the security level depends based on the service providers and the type of cloud clients use. Public cloud is better suited for business purposes for managing the load. This type of cloud is economical due to the decrease in capital overheads.

Advantages of the Public cloud are:

- Flexible
- Reliable
- High Scalable
- Low cost
- Place independence

Disadvantages of public cloud are:

- Less Secured
- Poor Customizable

3. Explain hybrid cloud mention its advantage and disadvantages.

Hybrid Cloud is another cloud computing type, which is integrated, i.e., it can be a combination of two or more cloud servers, i.e., private, public or community combined as one architecture, but remain individual entities. Non-critical tasks such as development and test workloads can be done using public cloud whereas critical tasks that are sensitive such as organization data handling are done using a private cloud. Benefits of both deployment models, as well as a community deployment model, are possible in a hybrid cloud hosting. It can cross isolation and overcome boundaries by the provider; hence, it cannot be simply categorized into any of the three deployments - public, private or community cloud.

Advantages of Hybrid Cloud Computing are:

- Flexible
- Secure
- Cost Effective
- Rich Scalable

Disadvantages of Hybrid Cloud Computing are:

- Complex networking problem
- Organization's security Compliance

4. Write the benefits of grid computing.

Grid computing is a computer network in which each computer's resources are shared with every other computer in the system.

Benefits of Grid computing are:

- It is a cost-effective way to use a given amount of computer resources.
- It is a way to solve problems that need a tremendous amount of computing power.
- The resources of several computers can be shared cooperatively, without one computer managing the other.
- In grid computing, a large project is divided among multiple computers to make use of their resources. Cloud computing does just the opposite.

5. Write the characteristics of SaaS.

- SaaS makes the software available over the Internet.
- The Software is maintained by the vendor rather than where they are running.
- The license to the software may be subscription based or usage based. And it is billed on recurring basis.
- SaaS applications are cost effective since they do not require any maintenance at end user side.
- They are available on demand.
- They can be scaled up or down on demand.
- They are automatically upgraded and updated.
- SaaS offers share data model. Therefore, multiple users can share single instance of infrastructure.
- Required to hard code the functionality for individual users.
- All users are running same version of the software.

6. What are the benefits of Multitenant nature of SaaS.**Benefits of SaaS multi-tenant architecture are:**

- Shared infrastructure leads to lower costs.
- Elasticity - the ability to scale up and down based on current requirements demanded by the business - is a benefit of multi-tenancy.
- Multitenancy feature distributes cost across a large pool of user via subscription.
- Ongoing maintenance and updates: Customers don't need to pay costly maintenance fee to keep their software up to date. Vendors roll out new features and updates. These are often included with a SaaS subscription.

(Questions for Understanding)

1. What do you mean by cloud computing? Write the uses of cloud computing?

Cloud computing refers to applications and services that run on a distributed network using virtualized resources and accessed by common Internet protocols and networking standards.

Uses of cloud computing are:

1. Cloud Follows Pay for What you Use Model: This means users will have to pay only for the amount of service used by them. This way, money spent on improving business through cloud doesn't prove to be expensive.
2. 24*7 Availability and Performance Oriented Infrastructure
3. Cloud Technology believes in Optimization of Existing Resources
4. Disaster Recovery Plan Help to Restore Data Quickly on Cloud
5. Agile Development on Cloud.
6. Cloud, Cost-Effective in Nature

The cloud promotes device independence, the cost of hardware and software automatically cuts short. Although, the initial expense of deploying cloud architecture is there, but that also focuses on pay for what you use model.

7. Cloud Architecture is Exceptionally Flexible

Cloud has various deployment models like private, public, hybrid, community which can control the access of control resources, hence flexible.

8. SRC (Security, Risk Management and Compliance Management)

2. Briefly explain about web 2.0 and its benefits.

Web 2.0 used to elaborate the second generation of www i.e. World Wide Web. It focused on the ability for user or people to collaborate and share information online via social media, web based communities, blogging and internet of course.

Benefits of Web 2.0 are:

- Access all the services available at internet anytime or from anywhere.
- Web services are user friendly or has better userinterface.
- Multiple kind or different type of media is available on web.
- It can build dynamic learning communities.
- Anyone can have real time discussion now.
- Success for search results is efficient.
- Updates of any information are easy.

3. Briefly explain all cloud deployment models.

There are four types of deployment models:

1. Public Cloud
2. Private Cloud
3. Hybrid Cloud
4. Community Cloud

Public Cloud

Public Cloud is a type of cloud hosting that allows the accessibility of systems & its services to its clients/users easily. Some of the examples of those companies which

provide public cloud facilities are IBM, Google, Amazon, Microsoft, etc. This cloud service is open for use. This type of cloud computing is a true specimen of cloud hosting where the service providers render services to various clients. From the technical point of view, there is the least difference between private clouds and public clouds along with the structural design. Only the security level depends based on the service providers and the type of cloud clients use. Public cloud is better suited for business purposes for managing the load. This type of cloud is economical due to the decrease in capital overheads.

Private Cloud

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Hybrid Cloud

Hybrid Cloud is another cloud computing type, which is integrated, i.e., it can be a combination of two or more cloud servers, i.e., private, public or community combined as one architecture, but remain individual entities. Non-critical tasks such as development and test workloads can be done using public cloud whereas critical tasks that are sensitive such as organization data handling are done using a private cloud.

Community Cloud

Community Cloud is another type of cloud computing in which the setup of the cloud is shared manually among different organizations that belong to the same community or area. Example of such a community is where organizations/firms are there along with the financial institutions/banks. A multi-tenant setup developed using cloud among different organizations that belong to a particular community or group having similar computing concern. Thus, community-based cloud users need to know and analyse the business demand first.

4. Explain briefly the cloud service models.

- **Infrastructure as a Service (IaaS):** Earlier called “Hardware as a service”
 - Service provider delivers a virtual computing resources(storage, networking, servers...) over the internet
 - Example: AWS (Amazon Web Services), GCP (Google Cloud Platform), firewall, Storage.
- **Platform as a Service (PaaS):** Service provider delivers a computing platform over the Internet.
 - users can build and deliver applications.
 - Example: Salesforce.com, Google App Engine.
- **Software as a Service (SaaS):** Service provider delivers software applications over the Internet.
 - Users subscribe to the software and access it via internet.

5. Write a short note on open SaaS.

Open SaaS refers to software as a service (SaaS) based on open source code. Similar to SaaS applications, Open SaaS is a web-based application that is hosted, supported and maintained

by a service provider accessed via internet. Open SaaS is best for community software. Open SaaS uses those SaaS applications, which are developed using open source programming language. These SaaS applications can run on any open source operating system and database.

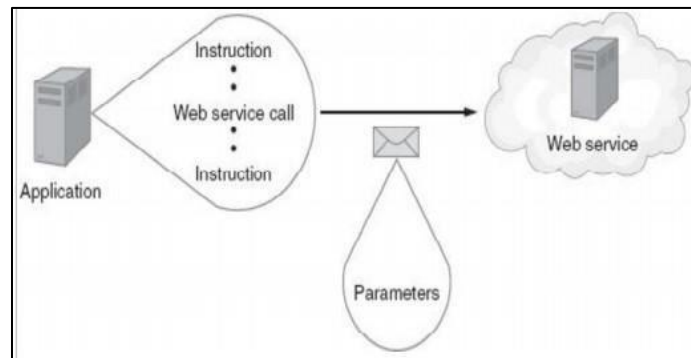
Open SaaS has several benefits listed below:

1. No License Required
2. Low Deployment Cost
3. Less Vendor Lock-in
4. More portable applications
5. More Robust Solution

6. Briefly explain Service Oriented Architecture.

Service-oriented architecture (SOA) is an application development methodology with which developers create solutions by integrating one or more web services. Think of a web service as a function or subroutine a program can call to accomplish a specific task. As shown in figure, when a program running on computer calls a web service, a message, possibly containing parameter values, is sent across the network (or Internet) to the computer housing the web service. That computer, in turn, performs its processing and normally returns a result to the caller.

Some developers refer to web services as remote-procedure calls. Further, developers refer to a set of web services as an application program interface (API). Amazon and eBay, for example, provide APIs that programmers can use to purchase products from across the web using the programs they create.



To call a web service, a program typically sends a message to the web service that resides on a remote computer and then waits for the web service to return a result. SOA architecture is viewed as five horizontal layers. These are described below:

- **Consumer Interface Layer:** These are GUI based apps for end users accessing the applications.
- **Business Process Layer:** These are business-use cases in terms of application.
- **Services Layer:** These are whole-enterprise, in service inventory.
- **Service Component Layer:** These are used to build the services, such as functional and technical libraries.
- **Operational Systems Layer:** It contains the data model.

Unit - II

Multiple Choice Questions

(Questions for understanding)

1. _____ is a form of cloud computing that provides virtualized computing resources .
A. PaaS
B. SaaS
C. **IaaS**
D. IDaaS
2. _____ offers browser-based development environment.
A. **PaaS**
B. SaaS
C. IaaS
D. IDaaS
3. Which service model provides access to computing platform over cloud computing solution.
A. **PaaS**
B. SaaS
C. IaaS
D. IDaaS
4. Event-based processor scheduling is an issue of _____
A. SaaS
B. **PaaS**
C. IaaS
D. IDaaS
5. _____ offers management of identity information as a digital entity
A. **IDaaS**
B. SaaS
C. PaaS
D. IaaS
6. The facilities that house cloud storage systems are called _____
A. Load balancing
B. Virtualization
C. **Data centers**
D. NAS
7. Expand SSO
A. **Single sign-On**
B. Server sign-On
C. Single sign-Of
D. Single server-On

8. Single sign-on is a service of _____
- A. SaaS
 - B. IaaS
 - C. PaaS
 - D. IDaaS**
9. comprises cloud-based solutions for identity and access management (IAM) functions.
- A. SaaS
 - B. IaaS
 - C. PaaS
 - D. IDaaS**
10. Cloud provider maintaining software versions and patch installation is a benefit of the _____ solution.
- A. Nas
 - B. PaaS**
 - C. IaaS
 - D. IDaaS
11. Knowledge ,Possession and inheritance is a criteria used in
- A. MIM(Mobile Id Management)**
 - B. MFA(Multi factor Authentication)
 - C. SSO(Single Sign On)
 - D. Open Id
12. _____ refers to set of attributes associated with something to make things recognizable.
- A. Object
 - B. Instance
 - C. Identity**
 - D. Class
13. AAA in Mobile Id Management is _____
- A. Access, Authorization and Accounting
 - B. Authentication, Available and Access
 - C. Authentication, Authorization and Available
 - D. Authentication, Authorization and Accounting**
14. Expand MFA
- A. Multi-factor authorization
 - B. Multi-factor authentication**
 - C. Multi-factor accounting
 - D. Multi -factor application
15. Dropbox is an example for _____
- A. IAAS solution
 - B. Cloud storage solution**
 - C. IDaaS solution

D. SAAS solution

16. MFA(Multi factor authentication) and biometric are the core components of
- A. SSO
 - B. open ID
 - C. **IDaaS**
 - D. cloud based NAS
17. An open standard that acts as a single user identification system which can be used across multiple websites.
- A. Multi Factor Authentication
 - B. **Open ID**
 - C. Single Sign On
 - D. Mobile ID Management MIM
18. Two factor authentication is used in
- A. Multi Factor Authentication
 - B. Open ID
 - C. Single Sign On
 - D. **Mobile ID Management MIM**
19. Amazon Web Services (AWS) and Google Cloud Platform (GCP) are examples of
- A. IDaaS providers
 - B. SaaS providers
 - C. PaaS providers
 - D. **IaaS providers**
20. Which of the following is the benefit of Infrastructure as a service
- A. **Reduced hardware cost**
 - B. Event based processor scheduling
 - C. Virtual machine sprawl
 - D. browser-based development environment
21. The process of distributing workloads & computing resources within a cloud technology's environment.
- A. **Cloud load-balancing**
 - B. Multi-factor authentication
 - C. Single Sign On
 - D. Network Attached Storage
22. _____ is an optimization technique which can be used to enhance utilization & throughput, lower latency, reduce response time & avoid overloading of systems.
- A. Redundancy
 - B. **Load balancing**
 - C. Virtualization
 - D. Scalability
23. In which system, a user log in to a system one time and then move freely among related servers and applications without having to authenticate him- or herself each time.
- A. MIM

B. **SSO**

C. MFA

D. PaaS

24. The scheduling algorithm ,round robin & weighted round-robin are mostly used to assign resource requested, by a_____

A. **load balancers**

B. virtual machine

C. NAS

D. SSO service

25. Which Algorithm is used by load balancer to assign requested resource

A. Shortest job first

B. Priority based

C. **Round robin**

D. First come first serve

Questions carrying 4 marks

(Questions for understanding)

1. **What do you mean by PaaS? Write the characteristics of PaaS.**

Platform as a Service (PaaS) is another service model which provides access to computing platform over cloud computing solution. PaaS supplies all the resources required to build applications and services completely from the Internet, without having to download or install software.

Characteristics:

Here are the characteristics of PaaS service model:

- PaaS offers browser-based development environment. It allows the developer to create database and edit the application code either via Application Programming Interface or point-and-click tools.
- PaaS provides built-in security, scalability, and web service interfaces.
- PaaS provides built-in tools for defining workflow and approval processes and defining business rules.
- It is easy to integrate with other applications on the same platform.
- PaaS also provides web services interfaces that allow us to connect the applications outside the platform.

2. **What do you mean by IaaS? Write the characteristics of IaaS.**

Infrastructure as a service (IaaS) is a form of cloud computing that provides virtualized computing resources on demand over the internet. Amazon Web Services (AWS) and Google Cloud Platform (GCP) are examples of independent IaaS providers.

Characteristics are:

Here are the characteristics of IaaS service model:

- Virtual machines with pre-installed software.
- Virtual machines with pre-installed Operating Systems such as Windows, Linux, and Solaris.

- On-demand availability of resources.
- Allows storing copies of particular data in different locations.
- The computing resources can be easily scaled up and down.

3. Write about disadvantage of PaaS.

- **Lack of portability between PaaS clouds**

Although standard languages are used yet the implementations of platforms services may vary. For example, file, Queue, or hash table interfaces of one platform may differ from another, making it difficult to transfer workloads from one platform to another.

- **Event based processor scheduling**

The PaaS applications are event oriented which poses resource constraints on applications, i.e., they have to Answer a request in a given interval of time.

- **Security engineering of PaaS applications**

Since the PaaS applications are dependent

4. Brief on cloud based NAS device.

NAS devices are hard drives that are connected to a network.

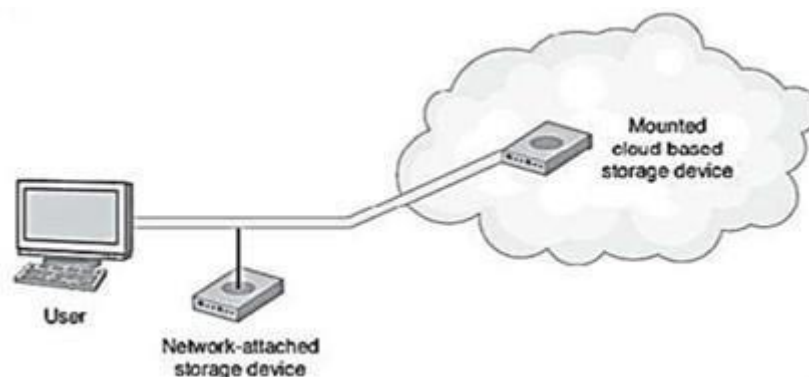
NAS is connected to an entire network, meaning that it can be accessed by any number of PCs as long as they are also on the same network.

NAS devices generally have some kind of built-in operating system which adds software functions like native media streaming, printer streaming, or remote access. But otherwise they functionally similar to any other kind of external drive you may have used before.

Cloud based NAS device which is present in cloud-based storage as a mounted device, may be replicated in the cloud to meet a company data redundancy needs.

Most NAS devices have software options that allow you to configure remote access (i.e. access from beyond your local network) so that you can get to your documents and files wherever you are in the world.

Cloud-based storage services Dropbox effectively let you rent space on someone else. Most NAS devices have software options that allow you to configure remote access (i.e. access from beyond your local network) so that you can get to your documents and files wherever you are in the world.



5. Write a short note on mobile ID management.

Identity management is essential to authentication, encryption and user activity policy. Mobile identity management is the natural evolution of what's historically been called AAA: authentication, authorization and accounting, three of the core components of any effective IT security strategy.

But identity management isn't just about wireless networking. The most important thing to remember when it comes to mobile identity management best practices is that the technology needs to be deployed across the entire network, no matter how a user connects.

Two-factor authentication: Because it's impossible to guarantee absolute security, each identity management element must be as reliable as possible. Mobile identity management systems that make use of the "something you have plus something you know" ethic of two-factor authentication have a major advantage. The second factor might use a specific device, biometrics, hardware token or other element.

With mobile identity and access management features, enterprises can control and manage users activity on company network and devices. MIM also helps businesses avoid the hassle of managing multiple usernames and passwords while improving end user experience.

6. Write a short note on cloud storage system.

There are hundreds of different cloud storage systems. Some have a very specific focus, such as storing Web e-mail messages or digital pictures. Others are available to store all forms of digital data. Some cloud storage systems are small operations, while others are so large that the physical equipment can fill up an entire warehouse. The facilities that house cloud storage systems are called data centers.

At its most basic level, a cloud storage system needs just one data server connected to the Internet. A client (e.g., a computer user subscribing to a cloud storage service) sends copies of files over the Internet to the data server, which then records the information. When the client wishes to retrieve the information, he or she accesses the data server through a Web-based interface. The server then either sends the files back to the client or allows the client to access and manipulate the files on the server itself.

(Questions for Application)

1. What are the advantages of IaaS.Explain?

Advantages of IaaS are:

Dynamic: Users can dynamically opt & configure devices such as CPU,storage,drive, etc.

- Easy Access: Users can easily access the vast cloud computing power.
- Renting: Flexible and efficient while renting IT infrastructure.

1. Full control of computer resources through administrative access to VM's.

- IaaS allows the consumer to access computing resources through administrative access.
- Consumer issues administrative command to cloud provider to run the virtual machine or to save data on cloud's server.
- Consumer issues administrative command to virtual machines they owned to start web server or installing new applications.
- Portability, Interoperability with Legacy Applications
- It is possible to maintain legacy between applications and workloads between IaaS clouds.

2. What do you mean load balancing? Write its application.

Cloud load-balancing is the process of distributing workloads & computing resources within a cloud technology's environment.

Application of Load Balancing

High performance level of tasks can be achieved using this load balancing technique on a lower cost than traditional on-premises load-balancing technology. In addition to workload and traffic distribution, cloud technology's load balancing can also provide health check for Cloud applications.

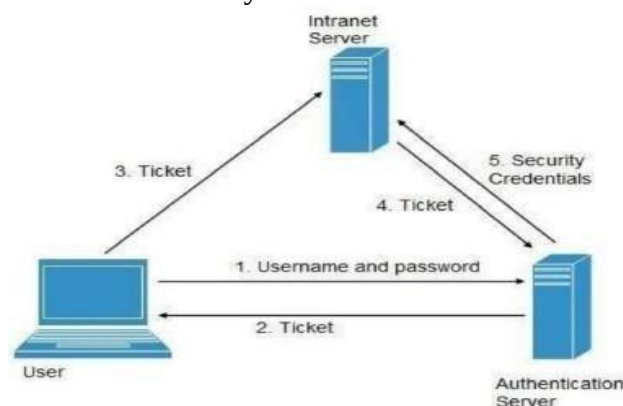
Load balancing is an optimization technique which can be used to enhance utilization & throughput, lower latency, reduce response time & avoid overloading of systems.

Without load balancing, it would have been very difficult to manage the cloud technology. It provides necessary redundancy procedures to make unreliable system reliable via managed redirection.

Load balancing system can use a different mechanism to assign direction to various services. In a simple case, load balancers are used to listen to network ports of the service request. When the request from a client arrives, load-balancers use the scheduling algorithm (round robin & weighted round-robin are mostly used) to assign where the request is sent.

3. What do you mean by SSO? Explain the working of SSO.

To solve the problem of using different username and password combinations for different servers, companies now employ Single Sign-On software, which allows the user to login only one time and manage the access to other systems.



Following steps explain the working of Single Sign-On software:

- User logs into the authentication server using a username and password.
- The authentication server returns the user's ticket.
- User sends the ticket to intranet server.
- Intranet server sends the ticket to the authentication server.
- Authentication server sends the user's security credentials for that server back to the intranet server.

4. What do you mean by open ID? Mention the benefits of Open ID.

OpenID is a unified user identification method released as an open standard that essentially acts as a single user identification system that can be used across multiple websites.

Benefits of Open ID are:

- Increased site conversation rates
- Access to greater user profile content
- Fewer problems with lost passwords
- Ease of content integration into social networking site

5. Explain benefits of Platform as a service.

1. Lower administrative overhead

Customer need not bother about the administration because it is the responsibility of cloud provider.

2. Lower total cost of ownership

Customer need not purchase expensive hardware, servers, power, and data storage.

3. Scalable solutions

It is very easy to scale the resources up or down automatically, based on their demand.

4. More current system software

It is the responsibility of the cloud provider to maintain software versions and patch installations.

7. Summarize how storage redundancy is beneficial in cloud computing.

Cloud storage systems generally rely on hundreds of data servers. Because computers occasionally require maintenance or repair, it's important to store the same information on multiple machines. This is called redundancy. Without redundancy, a cloud storage system couldn't ensure clients that they could access their information at any given time. Most systems store the same data on servers that use different power supplies. That way, clients can access their data even if one power supply fails.

Not all cloud storage clients are worried about running out of storage space. They use cloud storage as a way to create backups of data. If something happens to the client's computer system the data survives off-site. With 1000's of jobs to manage jam network interruption can be expected.

A WAN optimization services, reduces network latency, helps avoid blockage in network traffic and keeps the speed at optimum level.

A typical cloud storage system architecture includes a master control server and several storage servers.

Unit - III

Multiple Choice Questions

(Questions for Application)

1. Which of the following storage devices exposes its storage to clients as Raw storage that can be partitioned to create volumes?
 - A. **block**
 - B. file
 - C. disk
 - D. cloud

2. _____ means the storage is preconfigured for the customer. The customer can neither format, nor install his own file system or change drive properties.
 - A. Configured cloud storage
 - B. Managed Cloud Storage
 - C. Unformatted cloud storage
 - D. **Unmanaged Cloud Storage**

3. _____ offers online storage space on-demand.
 - A. **Managed cloud storage**
 - B. Partitioned cloud storage
 - C. Unmanaged cloud storage
 - D. Open cloud Storage

4. _____ is a strategy for sending a copy of a physical or virtual file or database to a secondary, off-site location for preservation in case of equipment failure or catastrophe.
 - A. **Cloud Backup**
 - B. Storage virtualization
 - C. Block Storage
 - D. Cloud database

5. _____ backups copy the entire data set every time a backup is initiated.
 - A. Incremental Backup
 - B. **Full Backup**
 - C. Differential Backup
 - D. hybrid backup

6. Which Back up service copy data from one cloud to another.
 - A. Point-in-time snapshots
 - B. **C2C back up**
 - C. Differential back up
 - D. Cloud seeding

7. _____ and continuous backups help, but costs rise as backup frequency increases.
 - A. **Point-in-time snapshots**
 - B. C2C back up

- C. Differential back up
 - D. Full back Up
8. _____ backups only back up the data that has been changed or updated since the last backup.
- A. **Incremental Backup**
 - B. Full Backup
 - C. Differential Backup
 - D. hybrid back up
9. Which is the common form of cloud backup used because of less resource use.
- A. **Incremental Backup**
 - B. Full Backup
 - C. Differential Backup
 - D. hybrid back up
10. Which of the following database is well-suited for unstructured data, such as social media content, photos and videos.
- A. MySQL
 - B. Oracle
 - C. IBM DB2
 - D. **NoSQL**
11. Which of the following is not a block storage options in the cloud.
- A. **AWS Elastic Backup Storage**
 - B. Azure Premium Storage
 - C. AWS Elastic Block Storage
 - D. Google Persistent Disks
12. Any application which requires service side processing like Java, PHP, and .Net will require _____
- A. **Block storage**
 - B. Network storage
 - C. File storage
 - D. Disk storage
13. RAID Volumes, where you combine multiple disks organized through striping or mirroring, uses which storage?
- A. **Block**
 - B. Network
 - C. File
 - D. Disk Drive
14. MySQL, Oracle, IBM DB2 and Microsoft SQL Server are Popular cloud platforms for _____
- A. Cloud Storage
 - B. **Cloud database**

- C. Cloud block storage
 - D. Cloud Provider
15. How many categories of storage devices broadly exist in cloud?
- A. one
 - B. **two**
 - C. three
 - D. four
16. _____ is a situation in which many under-utilized servers utilize more space or consume more resources than can be justified by their workload.
- A. **Server sprawl**
 - B. Server utilization
 - C. Server scheduling
 - D. Server Virtualization
17. A technique in which, cloud backup vendor sends a storage device such as a disk drive to its new customer, which then backs up the data locally onto the device and returns it to the provider.
- A. Incremental Back Up
 - B. Database virtualization
 - C. mirroring
 - D. **Cloud seeding**
18. _____ is technology that lets you create useful IT services using resources that are traditionally bound to hardware.
- A. Mirroring
 - B. Cloud back up
 - C. **Virtualization**
 - D. Cloud Storage
19. Multiple virtual servers can run on a physical server at the same time is called _____
- A. Mirroring
 - B. Sprawl
 - C. **Partitioning**
 - D. Isolation
20. The Virtual server running on the physical server are safely separated & don't affect each other is called _____
- A. Partitioning
 - B. **Isolation**
 - C. Encapsulation
 - D. Mirroring
21. _____ is the partitioning of physical server into several virtual servers and this division is mainly done to improve the utility of server resources.
- A. Mirroring

- B. Resource Virtualization
C. Server Partitioning
D. **Server Virtualization**
22. _____pools the physical storage from different network storage devices and makes it appear to be a single storage unit that is handled from a single console.
A. **Data storage Virtualization**
B. Network storage virtualisation
C. Server virtualization
D. database virtualization
23. Which of the following is a cloud storage service offered by IBM.
A. **Blue cloud**
B. Nirvanax
C. Gdrive
D. simple storage service(S3)
24. Which of the following is an open source cloud database platform.
A. NoSQL
B. **MySQL**
C. Oracle
D. MSAccess
25. Which of the following do not employ a table model.
A. **NoSQL**
B. MySQL
C. Oracle
D. MSAccess

Questions carrying 4 marks

(Questions for understanding)

1. Discuss the Pros and Cons of network storage.

Network Storage Pros and Cons are:

PROS:

- Provides an easy way for networked devices to access the same files
- It's easier to back up data for many users when they store their files in the same area
- Network storage saves local storage devices from reaching their capacity
- Less local storage needs to be purchased for users if they store most of it on the network

CONS:

- Devices not connected to the network cannot access the data (e.g. a down network prevents users from reaching their files)
- Serving shared files to users requires proper permissions to be set up, which can be time-consuming when lots of new users are configured at once

- More network storage requires more backup storage space

2. What is cloud based data storage. Explain its types.

Cloud Storage is a service that allows to save data on offsite storage system managed by third-party and is made accessible by a web services API.

Cloud storage can be broadly classified into two categories:

- Unmanaged Cloud Storage
- Managed Cloud Storage

Unmanaged Cloud Storage

Unmanaged cloud storage means the storage is preconfigured for the customer. The customer can neither format, nor install his own file system or change drive properties.

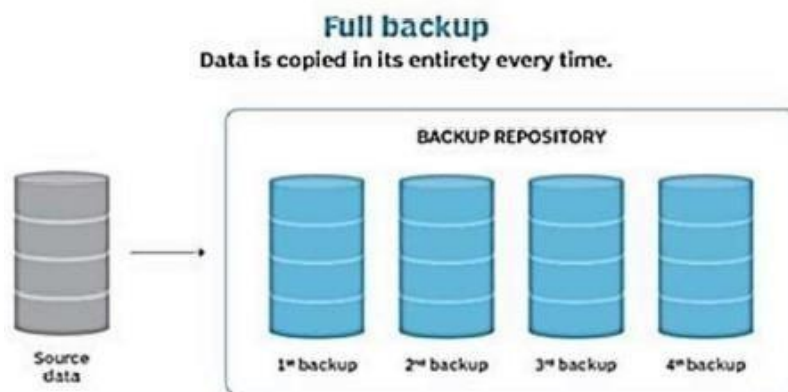
Managed Cloud Storage

Managed cloud storage offers online storage space on-demand. The managed cloud storage system appears to the user to be a raw disk that the user can partition and format.

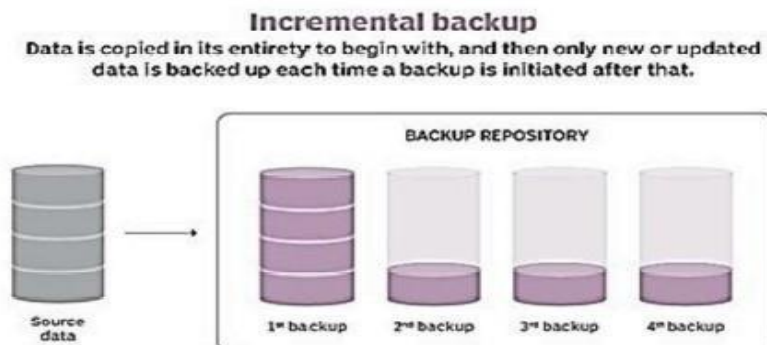
3. Explain the types of back up.

Types of backup:

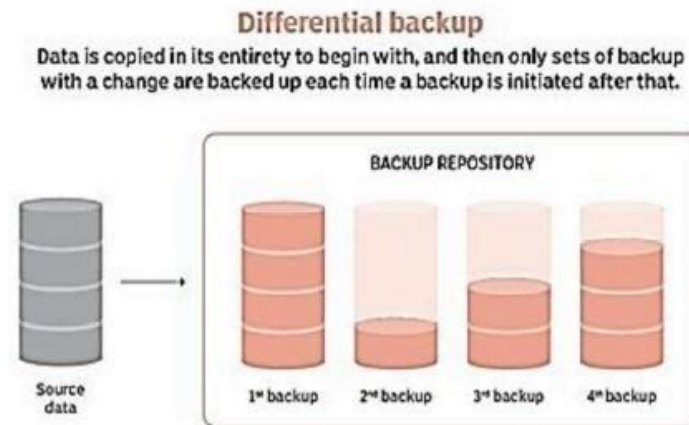
1. **Full backups** copy the entire data set every time a backup is initiated. As a result, they provide the highest level of protection. However, most organizations cannot perform full backups frequently because they can be time-consuming and take up too much storage capacity.



2. **Incremental backups** only back up the data that has been changed or updated since the last backup. This method saves time and storage space, but can make it more difficult to perform a complete restore.



3. **Differential backups** are similar to incremental backups because they only contain data that has been altered. However, differential backups back up data that has changed since the last full backup.



4. Explain the various cloud backup approaches.

Varieties of cloud backup approaches include:

- **Backing up directly to the public cloud**

One way to store organizational resources is by duplicating resources in the public cloud. This method entails writing data directly to cloud providers, such as AWS or Microsoft Azure. The organization uses its own backup software to create the data copy to send to the cloud storage service. The cloud storage service then provides the destination and safekeeping for the data, but it does not specifically provide a backup application.

- **Backing up to a service provider**

In this scenario, an organization writes data to a cloud service provider that offers backup services in a managed data center. The backup software that the company uses to send its data to the service may be provided as part of the service, or the service may support specific commercially-available backup applications.

- **Choosing a cloud-to-cloud (C2C) backup**

These services are among the newest offerings in the cloud backup arena. They specialize in backing up data that already lives in the cloud, either as data created using a software as a service (SaaS) application or as data stored in a cloud backup service. As its name suggests, a cloud-to-cloud backup service copies data from one cloud to another cloud. The cloud-to-cloud backup service typically hosts the software that handles this process.

- **Using online cloud backup systems**

There are also hardware alternatives that facilitate backing up data to a cloud backup service. These appliances are all-in-one backup machines that include backup software and disk capacity along with the backup server. The appliances are about as close to plug- and-play as backup gets, and most of them also provide a seamless (or nearly so) link to one or more cloud backup services or cloud providers.

5. Discuss the benefits of cloud database.

Cloud database offers the following distinct advantages:

- **Elimination of physical infrastructure** - In a cloud database environment, the cloud computing provider of servers, storage and other infrastructure is responsible for

maintenance and keeping high availability. The organization that owns and operates the database is only responsible for supporting and maintaining the database software and its contents. In a DBaaS environment, the service provider is responsible for managing and operating the database software, leaving the DBaaS users responsible only for their own data.

- **Cost savings** - Through the elimination of a physical infrastructure owned and operated by an IT department, significant savings can be achieved from reduced capital expenditures, less staff, decreased electrical and HVAC operating costs and a smaller amount of needed physical space.
- DBaaS benefits also include instantaneous scalability, performance guarantees, and failover support, declining pricing and specialized expertise.

6. Identify the features of Virtualization.

Features of virtualization are:

1. **Partitioning:** Multiple virtual servers can run on a physical server at the same time.
2. **Encapsulation of data:** All data on the virtual server including boot disks is encapsulated in a file format.
3. **Isolation:** The Virtual server running on the physical server are safely separated & don't affect each other.
4. **Hardware Independence:** When the virtual server runs, it can migrate to the different hardware platform.

7. What is storage virtualization explain its importance.

It pools the physical storage from different network storage devices and makes it appear to be a single storage unit that is handled from a single console.

Importance of Storage Virtualization are:

Storage virtualization is becoming more and more important in different forms such as:

1. Data is stored in a very convenient location. This is because if the host failure data don't get compromised necessarily.
2. By using storage level abstraction, it becomes flexible how storage is provided, protected, partitioned and used.
3. Storage Devices are capable of performing advanced functions such as disaster recovery, duplication, replication of data & re-duplication of data.
4. Storage virtualization facilitates management and monitoring of storage in a virtualized environment.
5. SAN Storage: SAN technology presents the storage as block-level storage & the storage is presented over the Ethernet network of OS.
6. File Server: OS writes the data to a remote server location to keep it separate and secure from local users.

(Questions for Application)

1. Write a short note on network storage.

Network storage is a term used to describe a storage device (usually many devices grouped together) that is available to a network. This kind of storage maintains copies of data across high-speed local area network (LAN) connections and is designed to back up files, databases, and other data to a central location that can be easily accessed via standard network protocols and tools.

Storage is an essential aspect of any computer. Hard drives and USB keys, for example, are

designed to hold personal data in a place that's close to where they need to access the information, like directly inside or next to their computer.

Network storage addresses these problems by providing a reliable, external data repository for all computers on the LAN to share efficiently. Freeing up local storage space, network storage systems also typically support automated backup programs to prevent critical data loss. Without a network storage solution, a file that needs accessing by multiple users who are not physically close would have to be emailed, moved manually with something like a flash drive, or uploaded online only to be downloaded again on the destination side. All of those alternative solutions pose time, storage, and privacy concerns that are alleviated with central storage. In basic terms, network storage is simply about storing data using a method by which it can be made available to clients on the network.

2. Write the advantages of cloud based data storage.

a. Accessibility and Synchronization

With local file storage, you only receive access to your data from a certain location. With cloud storage, on the other hand, the devices we now use every day, all day, become access points. This includes tablets, smartphones, desktops laptops, and more. With the device, you're able to access your data in any location with Internet access.

b. Collaboration Tools

Just its annoying to transfer files back and forth between devices, it's a pain to send dozens of emails just to share files. With cloud storage, there a better solution. Whatever you want to share, eliminating time wasted.

c. Cost Efficiency

Cloud storage eliminates the costs of using external hard drives or backup tapes, and the cost of time it takes to complete routing backups manually. Organization don't need to spend extra money on additional servers that are expensive to maintain or hire extra IT staff to do the work.

d. Space Efficiency

And the efficiency doesn't end with cost. When you move your data into cloud, it suddenly doesn't have a physical presence on your devices and hardware. It will no longer take up valuable space at home or in your office because the provider is the one offering the space to store it virtually.

e. Security and Recovery

Of course it can be frightening to store sensitive information in a new place, but the cloud has proven to be more secure than local business storage in most cases. With good storage solutions, data is encrypted in flight and at rest, unauthorized users are unable to access the files, and the cloud provider offers enterprise-grade security measures that businesses would otherwise not have.

3. Write the disadvantages of cloud based data storage.

Accessibility: If you have no internet connection, you have no access to your data.

- The network connection has to be very reliable.
- If the connection is slow or unreliable you might not be able to access your files.
- If the data storage company goes out of business then all of the data stored on their servers maybe lost (or not available to access).

Bandwidth & Cost issue: Several cloud storage services have a specific bandwidth allowance. If an organization surpasses the given allowance, the additional charges could be significant. However, some providers allow unlimited bandwidth. It usually costs money to store data in the cloud. Organisations might be charged for making large data backups or for retrieving large files from the cloud back to their network.

Data Security: There are concerns with the safety and privacy of important data stored remotely. You need to be certain that the company is reputable i.e. that they will perform

regular backups of your data and that they have good security in place so that no one can access your data

- You will need to remember a login and user name in order to access your data remotely.

4. Write a brief note on cloud back up.

Cloud backup, also known as online backup or remote backup, is a strategy for sending a copy of a physical or virtual file or database to a secondary, off-site location for preservation in case of equipment failure or catastrophe. The secondary server and storage systems are usually hosted by a third-party service provider, who charges the backup customer a fee based on storage space or capacity used, data transmission bandwidth, number of users, number of servers or number of times data is accessed. Cloud backup storage provides high encryption.

While there are multiple options and approaches to off-site backup, cloud backup serves as the off-site facility for many organizations. In an enterprise, the company might own the off-site server if it hosts its own cloud service, but the chargeback method would be similar if the company uses a service provider to manage the cloud backup environment.

Basic steps of cloud backup are:

- Files are designated for backup typically new or updated data.
- The backed - up data files are encrypted and sent off site to the cloud.
- Organization can choose between public cloud, private cloud or cloud to cloud backup methods.
- Files can be easily accessed and readily available in the event of a recovery.

5. Write a brief note on the concept of virtualization.

Virtualization is technology that lets you create useful IT services using resources that are traditionally bound to hardware. It allows you to keep physical machines full capacity by distributing its capabilities among many users or environments.

In more practical terms, imagine you have 3 physical servers with individual dedicated purposes.

One is a mail server, another is a web server, and the last one runs internal legacy applications.

Each server is being used at about 30% capacity just a fraction of their running potential. But since the legacy apps remain important to your internal operations, you have to keep them and the third server that hosts them, right?

Traditionally, yes. It was often easier and more reliable to run individual tasks on individual servers: 1 server, 1 operating system, 1 task. It was easy to give 1 server multiple brains. But with virtualization, you can split the mail server into 2 unique ones that can handle independent tasks to the legacy apps can be migrated. It's the same Hardware you're using more of it more efficiently.

6. Write a note on server virtualization and its uses.

Servers are computers designed to process a high volume of specific tasks really well so other computers like laptops and desktops can do a variety of other tasks. Virtualizing a server lets it to do more of those specific functions and involves partitioning it so that the components can be used to serve multiple functions.

Server virtualization is the partitioning of physical server into several virtual servers and this division is mainly done to improve the utility of server resource. In other words it is the masking of resources that are located in server which includes the number & identity of processors, physical servers & the operating system. This division of one physical server

into multiple isolated virtual servers is done by server administrator using software. The virtual environment is sometimes called the virtual private-servers.

In this process, the server resources are kept hidden from the user. This partitioning of physical server into several virtual environments; result in the dedication of one server to perform a single application or task.

This technique is mainly used in web-servers which reduces the cost of web-hosting services.

Instead of having separate system for each web-server, multiple virtual servers can run on the same system/computer.

The primary uses of server virtualization are:

- To centralize the server administration
- Improve the availability of server
- Helps in disaster recovery
- Ease in development & testing
- Make efficient use of server resources.

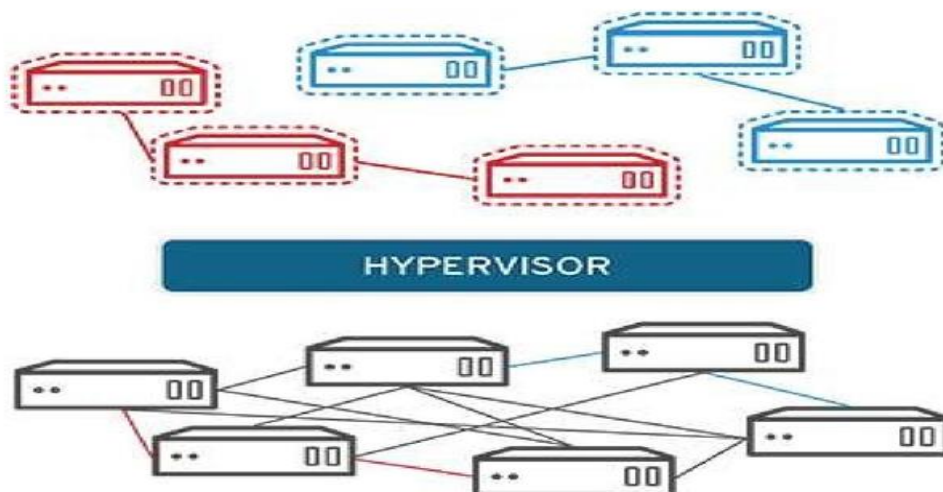
7. Write a note on virtual networking.

Virtual networking is a technology that facilitates the control of one or more remotely located computers or servers over the Internet. Data can be stored and retrieved, software can be run and peripherals can be operated through a Web browser as if the distant hardware were onsite.

Virtual networking facilitates consolidation of diverse services and devices on a single hardware platform called a virtual services switch. The centralization of control reduces the cost and complexity of operating and maintaining hardware and software compared with administering numerous separate devices in widely separated geographical locations.

Maintenance personnel and administrators can install device drivers, perform tests and resolve problems on the remote machines from a single location.

The chief limitation of virtual networking is the fact that certain problems can be resolved only by direct physical contact with the hardware involved. Examples include broken wires, frozen drives and defective chips. In some cases, virtual networking may make it possible to circumvent a damaged component by taking advantage of alternative resources until a technician can perform onsite repair or replacement. It may be necessary to install virtual networking software on the remote computers or servers to take advantage of this technology.



Unit - IV

Multiple Choice Questions

(Questions for understanding)

1. Cloud service providers keep their sites up to date to ensure that no vulnerabilities exist. This is possible because of?
 - A. Network performance
 - B. Patch management**
 - C. Resilience
 - D. Failback process
2. Which of the following is the hardest factor to determine?
 - A. Network performance
 - B. Network capacity**
 - C. Network delay
 - D. Network latency
3. Which of the following is a critical step in capacity planning?
 - A. Predict the future based on historical trends and other factors
 - B. Load the system until it is overloaded**
 - C. Deploy or tear down resources to meet your predictions
 - D. Measuring the work load
4. _____ measures the maximum amount of work that can be done using the current technology and then adds resources to do more work as needed.
 - A. Capacity planning**
 - B. SLA metric
 - C. Vendor lock in
 - D. Training procedure
5. The auditing process of a cloud-based solution can be of ____ types
 - A. two**
 - B. three
 - C. four
 - D. five
6. Which of the following model types is not trusted in terms of security?
 - A. Public**
 - B. Private

- C. Hybrid
 - D. Community
7. To keep essential functions up and running during a disaster and to recover with as little downtime as possible is the basic aim of _____
- A. Capacity planning
 - B. Business continuity planning**
 - C. Disk mirroring
 - D. Cloud seeding
8. What about vendor lock in not true in the following.
- A. When the Provider doesn't support data export of the customer
 - B. When the Provider service is not available elsewhere
 - C. the company can acquire and deploy the source code, put it on its own system and run it**
 - D. The agreement you sign with a cloud provider doesn't stipulate exit procedures
9. Which of the following is not correct about measures to avoid vendor lock in
- A. Negotiate both an entry and exit strategy upfront with your vendor
 - B. Have a backup vendor
 - C. Design portable applications
 - D. Agreement must not have exit procedures**
10. Which of the following is not a part of Providers security plan must not include
- A. Encryption policy.
 - B. Provider's multitenant use,
 - C. E-commerce processing
 - D. Un-authorization for data access**
11. Technology to maintain up-to-date copies of data in geographically dispersed locations the technology is _
- A. disk mirroring**
 - B. Capacity planning
 - C. Business continuity planning
 - D. Cloud seeding
12. _____ enables data access to continue uninterrupted if one location is disabled.
- A. disk mirroring**
 - B. Capacity planning
 - C. Disaster recovery
 - D. resilience
13. Business continuity has _____ key elements
- A. four
 - B. two
 - C. five
 - D. three**
14. Which of the following is not true about source code escrow agreement

- A. It places a copy of the provider's programming language source code with a third- party escrow company
 - B. It is a way to avoid the risk of vendor lock in
 - C. excuse the service provider from events beyond its control.**
 - D. if the solution provider fails, the company can acquire and deploy the source code, put it on its own system
15. SLA including a section detailing exclusions where agreement breach is not liable for punishments
- A. force majeure clause**
 - B. source code escrow
 - C. statement of objective
 - D. performance metrics

(Questions for skills)

16. What is the relation of capacity attribute to performance?
- A. same
 - B. different**
 - C. similar
 - D. Equal
17. Targeting the website by directing traffic from several sources to the end website is _____
- A. DDOS attack**
 - B. Vendor lock in
 - C. Ethical hacking
 - D. Web traffic
18. "Force majeure clause", which aims to
- A. excuse the service provider from events beyond its control.**
 - B. Punish the service provider from events beyond its control.
 - C. Force an action against customer
 - D. Force an action against provider
19. Which of the following is a security provider offering a range of cloud-based security solutions.
- A. McAfee**
 - B. MySQL
 - C. Rackspace
 - D. Cloud switch
20. Which mechanism refers to the process of transitioning user access from the thirdparty failover site back to the original production environment.
- A. Resilience
 - B. Fail back**
 - C. Contingency plan
 - D. Replication

21. What is the time taken for an issue to be resolved once logged by the service provider.
- A. Resilience time
 - B. Resolution time**
 - C. Response time
 - D. Reference time
22. What does an internal control of cloud solution refer to in auditing
- A. Policies and Procedure**
 - B. Auditing tools
 - C. IT staff
 - D. Bottlenecks
23. Resilience, Recovery and Contingency are the key elements of _____
- A. Capacity planning
 - B. Business continuity**
 - C. Cloud seeding
 - D. Disaster Recovery
24. The process of replicating and hosting servers and their data via a third party provider to enable a failover option in the case of an unexpected interruption.
- A. Disk mirroring
 - B. DRaaS**
 - C. DDOS
 - D. IDaaS
25. Replication, failover and fail back are the components of
- A. DRaaS Process**
 - B. Business continuity plan
 - C. Disk mirroring
 - D. Capacity planning

Questions carrying 4 marks

(Questions for understanding)

1. What do you mean by DDOS attack? How to avoid it.

Distributed Denial of Service (DDoS) attacks can result in hefty losses. Hackers target the website by directing traffic from several sources to the end website, and, as a result, the system gets overwhelmed. These DDoS attacks may tarnish the image of the company, as clients begin to lose trust.

A computer or networked device under the control of an intruder is known as a zombie, or bot. The attacker creates what is called a command – and - control server to command the network of bots, also called a botnet. The person in control of a botnet is sometimes referred to as the botmaster.

There are three types of DDoS attacks. Network-centric or volumetric attacks overload a targeted resource by consuming available bandwidth with packet floods. Protocol attacks target network layer or transport layer protocols using flaws in the protocols to

overwhelm targeted resources. And application layer attacks overload application services or databases with a high volume of application calls.

Cloud-based security systems guard this imminent threat with real-time scanning of potential risks; this function is further used as a warning tool for various systems which allows for the tracking of incoming threats and attacks instantly this enables website admins to divert the traffic to several different locations.

2. What do you mean by Patch management? Why is it required.

- ❖ **Security:** The most critical and obvious benefit of patch management is heightened network security. Patches are often created after a company has experienced a data breach to ensure other businesses' data remains safe, and applying a patch as quickly as possible lessens the risk of your business becoming affected. By securing your network before hackers exploit a program's security vulnerabilities, you can avoid the data theft, legal issues and lasting reputation damage that often follows a security breach.
- ❖ **Productivity:** A single patch can lead to a considerable boost in network productivity, and an entire patch management system can transform the efficiency of your entire company.
- ❖ **Remote protection:** Patch management programs can be implemented as part of a remote workforce support solution to protect all of your company's devices, no matter where in the world they're located.
- ❖ **Innovation:** Technology is moving quickly. It seems that just as soon as developers introduce one feature, a newer, better one comes along to take its place. patches — which usually contain new features and functionality—allow you to keep up with the latest technological innovations and improve your services without the need to buy entirely new software.

3. What is Business continuity? Explain its Key elements.

The most basic business continuity requirement is to keep essential functions up and running during a disaster and to recover with as little downtime as possible. A business continuity plan considers various unpredictable events, such as natural disasters, fires, disease outbreaks, cyberattacks and other external threats.

A business continuity plan has three key elements:

- Resilience
- Recovery
- Contingency
- A company can increase resilience by designing critical functions and infrastructures with various disaster possibilities in mind; this can include staffing rotations, data redundancy and maintaining a surplus of capacity. Ensuring resiliency against different scenarios can also help businesses maintain essential services on location and off-site without interruption.

- Rapid recovery to restore business functions after a disaster is crucial. Setting recovery time objectives for different systems, networks or applications can help prioritize which elements need to be recovered first. Other recovery strategies include resource inventories, agreements with third parties to take on company activity and using converted spaces for mission-critical functions.
- A contingency plan has procedures in place for a variety of external scenarios and can include a chain of command that distributes responsibilities within the organization. These responsibilities can include hardware replacement, leasing emergency office spaces, damage assessment and contracting third-party vendors for assistance.

4. What is SLA? What does it include.

A service-level agreement (SLA) is a contract between a service provider and its internal or external customers that documents what services the provider will furnish and defines the service standards the provider is obligated to meet.

SLA will typically include:

- a statement of objectives,
- Scope of services to be covered: a list of the services to be covered by the agreement
- define the responsibilities of the service provider and customer under the SLA.
- Performance metrics (response time , resolution time Penalties for contract breach /exclusions)



5. **What are the procedures to follow to ensure and audit system backup.**

Managers should consider different forms of backups. First, your company may back up user files from on-site computers to disks that reside within the cloud. Hopefully you will never require these backups; but regardless, you should periodically audit the backups, perhaps by checking that you can successfully restore randomly selected files of different users.

Second, if the cloud provider stores some or all of your company data, you must understand the provider's backup process (and include it in the SLA). For governance purposes, you should know if the data is encrypted, who has access to it, and if it is replicated to a remote facility. If it is backed up to another location, you must know where and how often.

Additionally, if the provider uses a database to store your company data, you need to know if and how the data is replicated and whether your company's information is stored in a private or multitenant database. You should also know the system's guaranteed uptime.

6. **What is vendor lock in? How to avoid it.**

Relationships can go bad even those with a cloud-solution provider. The agreement you sign with a cloud provider should stipulate exit procedures in case the provider fails to meet the service levels or breaches any other aspect of the contract.

IT managers must be able to control their company's data. In the event of a worst-case scenario,

a manager must be able to export the company data, ideally to a file that can be imported by another provider. Managers should test this capability before a problem arises.

Vendor lock-in occurs when a provider does not support data export or when a provider's service is unavailable through others. Thus, the customer is "locked in" to the relationship with the vendor. If this happens, a company may have to put up with breaches of its SLA because it has no other place to move its data. Managers should consider the risk of vendor lock-in before they enter into a provider agreement.

(Questions for Application)

1. **Write the security advantages of cloud Based solution.**

Data Encryption: Cloud-based systems give the required freedom to companies to choose their users who will be accessing the data that has been outsourced to the cloud. This way, any attempts to tamper with personal or profession data can be opposed.

Most companies face the threat of internal data theft by their employees, and stronger access controls can nip these threats in the bud. The multi-layered security features weed out the possibilities of a breach of data to a great extent. Data, irrespective of its type, needs to be protected at all times. Any violations can be hazardous to the goodwill and the functioning of an enterprise.

Avoid DDoS Attacks: Distributed Denial of Service (DDoS) attacks can result in hefty losses. Hackers target the website by directing traffic from several sources to the end website, and, as a result, the system gets overwhelmed. These DDoS attacks may tarnish the image of the company, as clients begin to lose trust.

There are three types of DDoS attacks. Network-centric or volumetric attacks overload a targeted resource by consuming available bandwidth with packet floods. Protocol attacks target network layer or transport layer protocols using flaws in the protocols to

overwhelm targeted resources. And application layer attacks overload application services or databases with a high volume of application calls.

Cloud-based security systems guard this imminent threat with real-time scanning of potential risks; this function is further used as a warning tool for various systems which allows for the tracking of incoming threats and attacks instantly this enables website admins to divert the traffic to several different locations.

Regulatory Compliance: Cloud computing security solutions usually provide reliable SOC1 and SOC2 certifications to the entertainment businesses. These certifications ensure periodic scrutiny of data and all types of possible glitches. Cloud-based solutions manage the requisite infrastructure for regulatory compliance and the protection of data. Detailed AWS reports about management of security controls ensure all organizations focus on their business operations, without worrying about compliance requirements

Secure Storage

Traditional storage solutions don't provide any protection against possible disasters that have the potential to erase required data from devices. Cloud computing allows the users to store their data safely, thereby negating any mishaps that may affect the equipment. Cloud storage solutions offer private, public, and hybrid solutions which the businesses may choose as per their requirements. The hybrid cloud storage systems allow the users to keep their data secure in the most effective manner.

Patch Management: The vulnerabilities of a website are often exploited by hackers to breach the security system of a company. Cloud service providers keep their sites up to date; further on, they ensure that no vulnerabilities exist. Moreover, cloud solutions offer real-time assistance to clients by providing companies with the option to scale cloud solutions during high traffic situations. This flexibility allows companies to reduce their cost of services substantially. These large number of security features are quite flexible, agile, and affordable. Enhanced security features offer sufficient protection to the private and financial data of both media and entertainment companies and help to protect data and intellectual property breaches.

2. Explain Disaster Recovery and components of DRaaS.

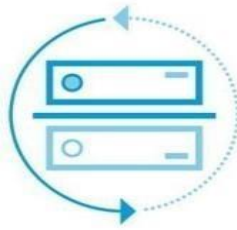
Disaster recovery is mainly data focused, concentrating on storing data in a way that can be more easily accessed following a disaster.

Disaster recovery planning specifies an organization's planned strategies for post-failure procedures.

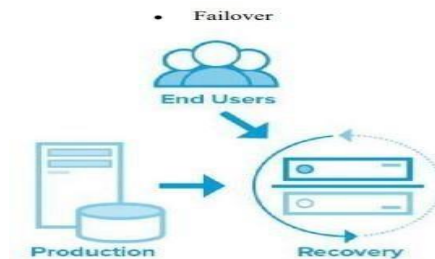
At the most basic level, DRaaS is the process of replicating and hosting servers and their data via a third party provider with the purpose of enabling a failover option in the case of an unexpected interruption or event.

- **Replication** is the process of duplicating data from a production environment and transmitting it to the third party host. Since most organizations have not fully-virtualized their infrastructure, the replication technology should have the capability to frequently capture both virtualized and non-virtualized (physical) servers. This is often referred to as a hybrid solution in instance where data is constantly added and

changed frequent data snapshots are essential in order to prevent data loss during failover.



- **Failover** is the transitioning of end-user access during an event to the third party hosting environment. A DRaaS solution should provide a simple, straight-forward process for declaring and failing over to the replicated environment. The speed in which the solution can “stand up” the replicated environment corresponds directly to the impact on business operations. Integrated tools and support should be in place so that failover can occur, even if your IT is unavailable.



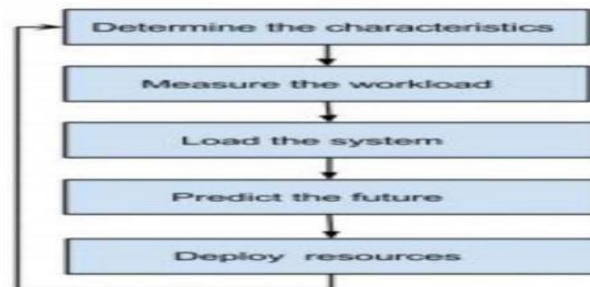
- **Failback** is the process of transitioning user access from the third party failover site back to the original production environment. Once the impacted environment is fully up and running again, the DRaaS solution should have processes and tools in place that allow for a seamless switch back to production. Immediately following failback, replication processes should be re-engaged to ensure continuous protection.



3. Summarize capacity Planning in cloud management.

For SaaS solutions, the cloud-solution provider will scale the site to match user demand. An IT manager, however, must define in advance key response - time metrics the solution must provide and then include those measures within the SLA.

For PaaS and IaaS solutions, the IT manager must initially estimate the solution's capacity plan, which defines the resources the solution will need to operate satisfactorily. The IT manager should also estimate the site's potential growth and define, with the help of the solution provider, the plan for scaling the site resources as well as the related costs.



Capacity planning measures the maximum amount of work that can be done using the current technology and then adds resources to do more work as needed. Several sites within the cloud provide system-performance reports that managers can use to measure current performance and the potential system benefit from scaling specific resources.

4. Summarize Auditing/inspection of a cloud based solution.

The company should examine the sources from which the values are derived to ensure that each is accurate and free from fraud. This inspection process is known as auditing. The auditing process can be internal (done by the company) or external (done by a third party), as shown in figure below

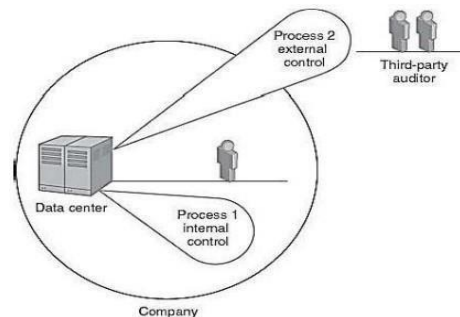


FIGURE Companies must audit the source of the values they measure and report using internal or external auditors.

- Internal audit: Conducted within an organization, they are voluntary. To be used as quality control checks, scrutinizing business operations against set standards. These standards can be governmentally set by regulatory bodies, or by the organization itself.
- External audit: Conducted externally to a business, by an independent, qualified auditor. These are NOT voluntary. They aim to make sure records kept by a business are complete, accurate and adhere to legal standards.

5. What is the use of monitoring the log files.

To identify potential system bottlenecks, detect errors within the system, and identify system resource use, the IT manager may examine various system log files. In a PaaS or

IaaS solution, the manager can likely turn on the log file reporting that meets his or her needs. For a SaaS solution, the manager should discuss in advance with the cloud service provider the various logs that will be available and the costs of running them, both in terms of dollars and system performance.

6. Write a note on SLA performance metrics.

SLAs establish customer expectations with regard to the service provider's performance and quality in a number of ways. Some metrics that SLAs may specify include the following:

- Availability and uptime percentage -- the amount of time services are running and accessible to the customer. Uptime is generally tracked and reported on per calendar month.
- Service provider response time -- the time it takes the service provider to respond to a customer's issue or request. A larger service provider may operate a service desk to respond to customer inquiries.
- Resolution time -- the time it takes for an issue to be resolved once logged by the service provider.
- The schedule for notification in advance of network changes that may affect users.
- Usage statistics that will be provided.

UNIT – V

Multiple Choice Questions

(Questions for Application)

1. Which of the following factors needs to be considered while moving from smartphone execution to a virtual machine running in the cloud?
 - A. Network bandwidth
 - B. Memory
 - C. Processing power
 - D. Storage
2. In which of the following computing, mobile device serves as the presentation platform or the display?
 - A. Soft
 - B. Mobile
 - C. API
 - D. Application
3. Which of the following feature is highly emphasized in open system for mobile application?
 - A. scalability
 - B. integrity
 - C. interoperability
 - D. readability

4. Which of the following has made the development of mobile Web applications much easier and more powerful?
- A. **Web Services**
 - B. Cloud Computing
 - C. IM
 - D. Analog communication
5. Which of the following is a push mechanism for publishing Web site data that is device-specific?
- A. **WAP**
 - B. GPRS
 - C. GAP
 - D. SAP
6. The process of moving data, applications or other business elements to a cloud computing environment is?
- A. **Cloud migration**
 - B. Cloud seeding
 - C. Business continuity Plan
 - D. Cloud bursting
7. Which is the feature that ensure the authorization of the user to access the server from any where regardless of device.
- A. **IP whitelisting**
 - B. Automation
 - C. Scaling
 - D. Migration
8. Which is a way to reduce the risk of vendor lock-in
- A. migration
 - B. automation
 - C. **open solutions**
 - D. IP whitelisting
9. The application expands into the cloud temporarily, When the on-site application encounters increased user demand, this process is known as _____
- A. Cloud migration
 - B. **Cloud Bursting**
 - C. Cloud switching
 - D. Cloud evolution
10. When the user demand declines, the application leaves the cloud, this is possible because of _____
- A. Cloud migration
 - B. **Cloud Bursting**

- C. Cloud switching
- D. Cloud scaling

11. What does demand period indicate as a performance indicator
- A. The cost going high because of excess application use
 - B. **The application have time frame when its usage is high or low**
 - C. The vendor scheme of billing the application as per extensive use
 - D. The application demand increasing strategy to be observed by user

(Questions for skills)

12. Access control, Auditing and Authentication are the key mechanism for?
- A. **protecting data**
 - B. business continuity
 - C. cloud switching
 - D. capacity plan
13. Which of the following are essential characteristic of smart phone in cloud ?
- A. A smart phone offers advanced calling features such as video calls or conferencing
 - B. **A smart phone has a recognizable operating system**
 - C. A smart phone comes with a touch screen
 - D. All of the mentioned
14. Mobile Lite is an extension of the _____ cloud service.
- A. **Salesforce.com**
 - B. Google.com
 - C. aol.com
 - D. yahoo.com
15. Which of the following is most popular open-system alternative to system like iOS or BlackBerry OS?
- A. **Android OS**
 - B. Symbian
 - C. Solaris
 - D. windows
16. Which of the following was built to serve as a mobile platform for Internet computing?
- A. Windows Phone
 - B. **Android**
 - C. Symbian
 - D. Microsoft
17. Difference between an app and widget is
- A. Widgets can not be downloaded, app's can be
 - B. Apps are mobile device independent ,widgets are not
 - C. Apps are always running ,widgets are not

D. **Widgets are always active and apps are device specific programs**

18. What did First-generation phones supported
- A. **analog communication**
 - B. digital communication.
 - C. Internet communication
 - D. Wifi connection
19. Digital communication was introduced in which cell phone generation
- A. first
 - B. **second**
 - C. third
 - D. fourth
20. Which of the following is a not a cloud migrationplatform
- A. RIGHTSCALE
 - B. 3TERA
 - C. **McAfee**
 - D. CLOUDSWITCH
21. The idea of mobile computing has been started since
- A. 2000
 - B. **1990**
 - C. 1980
 - D. 1973
22. Which of the following is not a part of Mobile cloudecosystem
- A. Apps and widget
 - B. **Compact disks**
 - C. Transmission speeds for upload and download operations
 - D. Keyboard interface
23. Abbreviate HIPAA act for health care specifying privacy requirements.
- A. Health infrastructure Privacy and Portability
 - B. Help Insurance Privacy and Portability
 - C. **Health Insurance Privacy and Portability**
 - D. Health information Privacy and Portability
24. Abbreviate FERPA act for education which specify privacy requirements
- A. Family Enterprise Rights and Privacy
 - B. Federal Educational Rights and Privacy
 - C. **Family Educational Rights and Privacy**
 - D. Forest Educational Rights and Privacy
25. Which changes the document content, such as changing a PNG graphic to a GIF graphic to improve the download performance or device display.
- A. **transcoder**
 - B. Widgets

- C. Smart phone
- D. W3C Mobile Checker

Questions carrying 4 marks

(Questions for Skills)

1. What do you mean by cloud migration? Explain the benefits of cloud migration.

The process of taking an application to the cloud, known as cloud migration.

Cloud Migration Benefits:-

i. Scalable Resources

The most visible benefit of cloud migration is the inherent feature of flexible server resources. This is the main reason why applications that require varying levels of server resources opt for the scalable server solutions offered by cloud hosting solutions.

ii. Cost Reduction

In an era when the competition has become fierce, every business is looking for cost-effective solutions to remain profitable. Cloud migration helps companies to reduce both operational and capital expenses simply because of the Pay – As – You - Go billing model that is ideal for businesses who now have to pay only for the resources their server consumes.

iii. Mobility

Since a cloud server is accessible from anywhere, users experience the versatility of access and ease of use like none other. IP whitelisting feature ensures the authorization of the user to access the server from any part of the world regardless of device. This function is commonly available in all PaaS cloud services, such as Cloudways.

iv. Automation

A known cloud migration problem is to keep the backend server to (managed) cloud migration, application management team do not have to worry about API and application update.

v. Flexible Operations

Cloud Solution allows organizations to deploy and test applications quickly over the cloud. Instead of manually using the application, the development teams can deploy the applications through automated deployment options without worrying about compatibility and performance issues.

2. Analyze a case study with reference to cloud migration.

CASE STUDY 1:

CLOUDSWITCH CLOUD MIGRATION

Many companies have enterprise-based applications that are widely used by their employees. These applications, therefore, are mission critical. Cloud Switch provides a downloadable application that companies can install their data center that securely maps the company's on-site applications to a cloud-based solution in a matter of minutes. In other words, CloudSwitch, provides a way for companies running Windows or Linux solutions to migrate quickly to the cloud. The CloudSwitch migration program requires no programming or development. Additionally, CloudSwitch provides a suite of cloud-

management tools that the IT staff can use to manage the solution after it moves to the cloud.

CASE STUDY 2:

3TERA CLOUD SOLUTIONS

Formerly known Computer Associates, CA Technologies provides a wide range of services and solutions to companies migrating to the cloud. To drive its cloud offerings, CA Technologies acquired 3Tera, a company that helps businesses move solutions to public and private clouds using a graphical user interface (GUI) tool. The 3Tera website offers a turnkey cloud computing platform, driven by the company's AppLogic software which allows large (enterprise) or small companies to migrate to the cloud through the use of the 3Tera virtual appliance, a device that behaves very much like a virtual machine. As user demand for an application grows, 3Tera can easily scale solutions.

3. What are the characteristics of application which indicate its performance in cloud?

- **Demand periods:** Does the application have periods of high or low demand. such as 8 A.M. to 5 P.M.?
- **Average users:** How many users typically use the systemsimultaneously?
- **Disk-storage requirements:** What are the application's typical disk-storage needs? Are the files permanent or temporary? Are most operations read or write operations?
- **Database-storage requirements:** What are the application's database requirements? Is the database replicated in real time? What is the application's database read/write ratio?
- **RAM use:** What is the application's range of physical and virtual memoryuse?
- **Bandwidth consumption:** What is the application's bandwidth requirement?
- **Caching:** How does the application currently cache data?

4. What are the IT governance issues to be considered for applicationmigration?

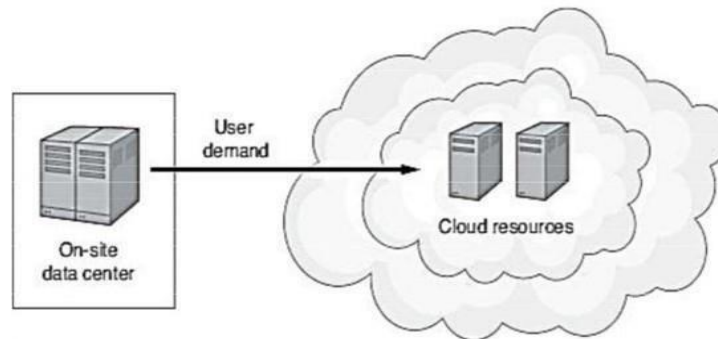
Before you migrate an application to the cloud, consider the following governance requirements:

- Identify how the cloud solutions align with the company businessstrategy.
- Identify and define the internal and external controls the company will need within the application, and at what control points, in order to validate that the application is performing correctly and is free from possible external modification.
- Describe risks the IT staff is trying to mitigate and ways the cloud can help.
- Describe who within the company will have access to data within the cloud and how they will get it.
- Determine who within the cloud provider's organization will have access to data within the cloud and how they will get it.
- Discover which performance - monitoring tools are available for your use.
- Determine how and when the cloud provider performs system updates and patches.

- Discover how the cloud provider logs errors and system events and how you can access them.

5. Summarize cloud bursting with diagram.

One of the clouds biggest advantage is its ability to scale on the fly to meet user demand. Some companies that run on-site applications have started to use the cloud as a way to scale their applications on demand. When the on-site application encounters increased user demand, the application expands into the cloud through a process called cloud bursting, as shown in FIGURE. When the user demand declines, the application leaves the cloud. Cloud bursting is most common for seasonal demand, or event-driven demand, such as the load on Google Maps when an earthquake or other natural disaster occurs.



6. What are the necessary measures to be taken to protect data before application migration?

Before you begin your application migration to a cloud provider, make sure that you back up your data so that you can revert, if necessary, to a known restore point. Then, be sure that you and the provider agree to the backup procedures that will be employed after they take control of your data. As discussed, it is easy to move a solution to a cloud provider. You need to ensure that it is equally easy to move out of the cloud if necessary.

To reduce the employee stress during an application's migration to the cloud and to increase employee productivity with the cloud-based tools, you should consider training before, during, and after the cloud migration. As you define your training requirements, consider the following:

- Employee preparedness for the SaaS solution
- Developer training on the solution application program interfaces (APIs)
- Administrator training for cloud-based operations
- IT-audit group training for corporate governance issues and internal controls
- Help desk support preparedness training
- Business continuity and disaster preparedness training

7. Debate on mobile cloud eco system.

An ecosystem is an environment that consists of living and non-living things with

which one interacts. Many cloud- based companies use the term ecosystem to describe the users environment. To that end, you might describe the mobile-cloud ecosystem as consisting of the following:

- Phone calls, which may be voice or face-to-face
- Web browsing
- Apps and widgets
- Voice commands and voice recognition
- Display screens
- Transmission speeds for upload and download operations
- Keyboard interface
- Touchscreens

8. Debate on evolution of mobile computing.

If you list inventions that have had the biggest impact on society, from the automobile to airplanes to television and the telephone, the cell phone may (or likely eventually will) top the list. It is important to note that in less than 30 years the cell phone has evolved from a heavy cumbersome device, to a small handheld device with more than 1 billion users. Today, the liveliest sector of the mobile phone market is smartphones, which integrate computing capabilities, and often a web browser, which allows the phones to provide a wide range of solutions. Beyond traditional phone calls, users use smartphones to do the following:

- Browse websites, including Google, Facebook, eBay, and more
- Place face-to-face video calls to phones and computers
- Perform GPS-based navigational operations
- Exchange text messages
- Perform e-commerce operations
- Run a myriad of applications (apps)

9. Illustrate the cell phone capabilities in terms of generations.

When you discuss phone capabilities, you will hear terms such as 3G and 4G. In the simplest sense, the G stands for generation. A 4G phone, therefore, is a fourth-generation phone. Although there are standards with respect to the potential speeds associated with each generation, the speed of most devices today depends upon the location (some cities support faster speeds), the provider network, and the phone technology, as well as whether the user is moving or stationary. TABLE describes the key generational attributes.

THE COMMON GENERATION OF CELL PHONES

Generation	Capability
1G	First-generation phones supported analog communication.
2G	Second-generation phones introduced digital communication.
3G	Third-generation phones supported faster speeds, which, in turn, made web browsing and e-mail readily available.
4G	Fourth-generation phones support near Wi-Fi speed, which enables rich media and video streaming.

10. Debate on Mobile players.

To understand mobile data communication, you should understand the players. To start, the operator, or network, is the company that makes the mobile network available. Within the United States, mobile operators include Verizon and AT&T. In the simplest sense, the operator owns the cell tower through which the data communication occurs. Most operators will provide internal groups that support developers in bringing mobile solutions to the market.

Within most mobile networks, operators will place special servers, called transcoders, which examine the content a mobile device is downloading from a website. The transcoder, in turn, may change the document content, such as changing a PNG graphic to a GIF graphic to improve the download performance or device display. If you are developing mobile content, you must be aware of how the transcoder may change the content. Unfortunately, standards for transcoders are still being developed. As a result, you may see differences, in some cases considerable ones, among the content produced by different transcoders.

Many operators may include proxy servers (proxies) which perform operations on behalf of a device. By serving as an intermediary, a proxy provides a level of security that separates the device from the web server with which the device is interacting.

11. Illustrate different ways to build mobile solutions?

When developers build mobile solutions, they can approach their solution in one of three ways:

First, they can build a web page, ideally targeted for a mobile display. As users browse the Web using mobile browsers built into their phones, the contents of the web page appear.

Second, developers can build an app, which typically is a device-specific program that users download and install (either free or for a price) onto a device. The app, in turn, displays an icon on the device, which users click to start the app.

Third, developers can create a widget, which is much like an app that the user downloads and installs to his or her mobile device. A widget differs from an app in that the widget is always Active. A clock widget icon, for example, might constantly display the current time, a weather widget icon, in turn might display changing weather conditions and a map widget may constantly update its icon to show the user location. In other words, a widget is always running and may possibly be communicating. Having many active widgets at one time might affect a device's performance.

12. Describe the development consideration for designing Mobile cloud solution.

If you work on a mobile solution, remember the following development considerations:

The mobile web is not the traditional web: You should not expect to use your traditional web pages as mobile solutions. You should optimize your web solutions and then optimize your mobile solutions.

Fast is good: Mobile data communication is still slower than most computer-based data communication solutions. As such, you should optimize your mobile web layout and design to maximize download performance.

Remember your goals and requirements: As you design your mobile solutions, keep your original goals and requirements in mind to ensure that your solution matches your business strategy.

You cannot support everything: Pick your largest market segment (or device) and focus your initial efforts there.

Do not treat mobile content as an afterthought: create, do not convert, mobile content: Do not simply convert your traditional web content for use on the mobile web. Instead, design your mobile content for optimal performance and market impact.

Handle different display sizes differently: Mobile applications should query the browser or device to determine the supported display size and then provide matching display content dynamically.